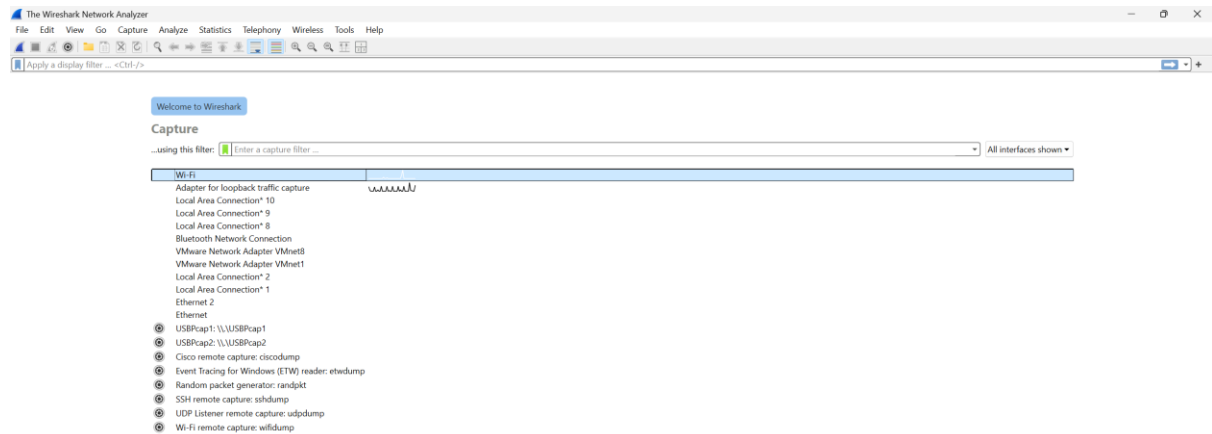


Aim: Using Wireshark, network analyzer, set the filter for ICMP, TCP, HTTP, UDP, FTP and perform respective protocol transactions to show/prove that the network analyzer is working.

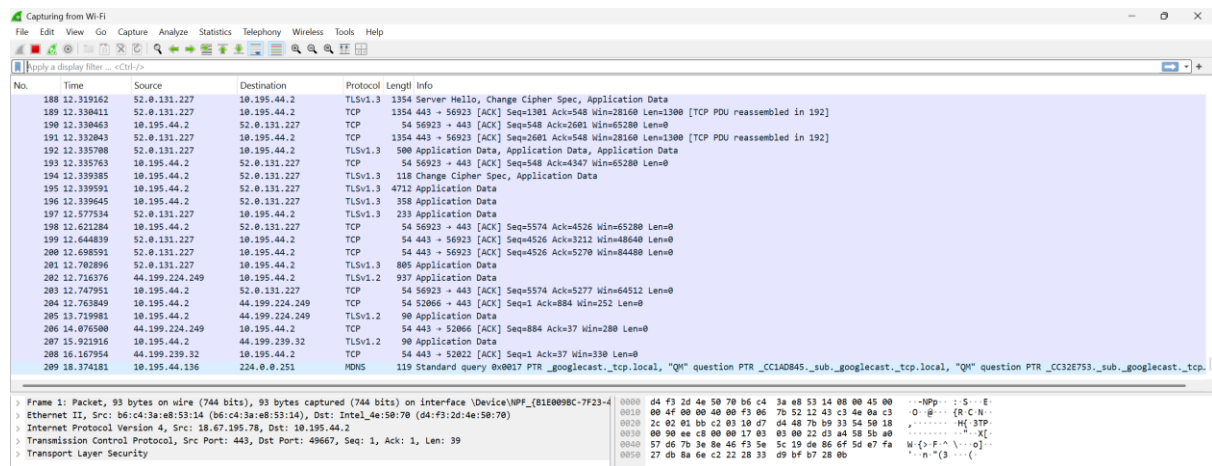
1. Opening Wireshark.

OUTPUT:



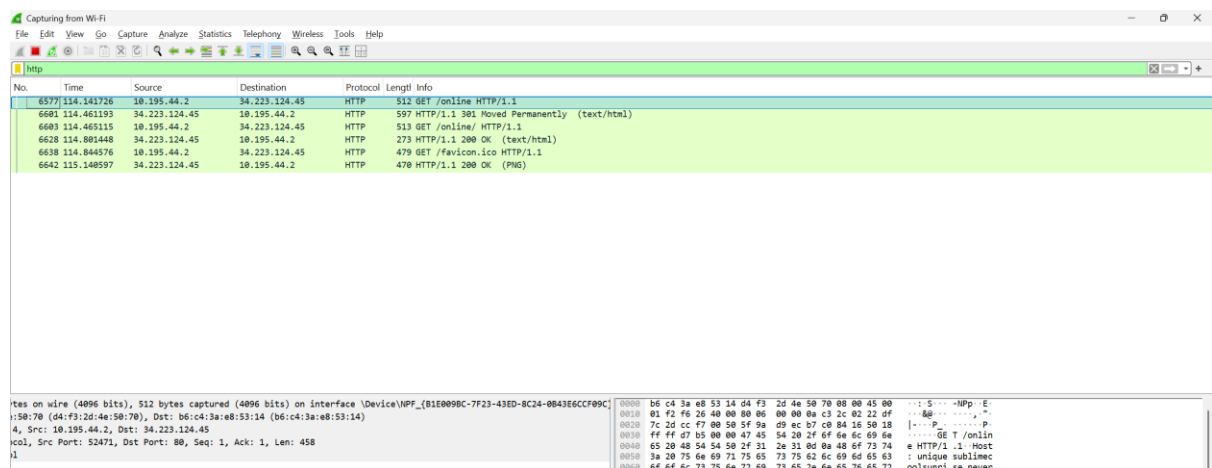
2. Capturing packets and traffic from Ethernet/WiFi.

OUTPUT:



3. Browsing a HTTP website: <http://neverssl.com>. Applying the HTTP filter to filter out all the HTTP-only traffic and packets.

OUTPUT:



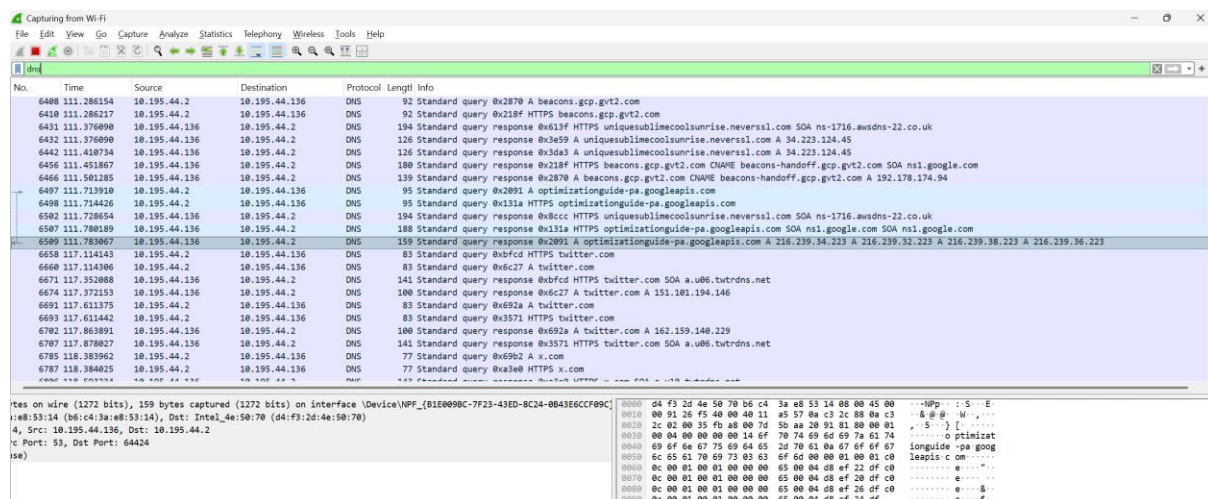
OUTPUT:



OUTPUT:



OUTPUT:



7. Applying the ICMP/ICMPv6 filter to capture the ICMP/ICMPv6 traffic and packets.

OUTPUT:

The image displays two screenshots of the Wireshark network protocol analyzer. The top screenshot shows an ICMP capture with two packets. The bottom screenshot shows an ICMPv6 capture with multiple Neighbor Solicitation and Neighbor Advertisement packets, followed by a detailed view of a selected packet.

Top Screenshot: ICMP Capture

No.	Time	Source	Destination	Protocol	Length	Info
8636	140.859075	10.195.44.136	10.195.44.2	ICMP	103	Destination unreachable (Network unreachable)
8677	141.145998	10.195.44.136	10.195.44.2	ICMP	86	Destination unreachable (Network unreachable)

Packet 8677 details:

- Frame 5: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{81E0898C-7F23-43ED-8C24-0B43E6CCF895}, Interface 0, ID 0
- 4, Src: 10.195.44.136, Dst: 10.195.44.2
- Protocol

Hex dump and ASCII view:

```
0000  d4 f3 2d 4e 50 70 b6 c4 3a e8 53 14 81 00 00 00  --Np...: $....
0010  00 00 45 c0 00 55 da 4c 00 00 40 01 31 8c 0a c3  --E..U..@..:..
0020  2c 88 0a c3 2c 02 03 00 09 5a 00 00 00 45 00    --...Z...E...
0030  00 39 38 60 40 00 80 11 b6 30 0a c3 2c 02 8e fa  --@...@...:..
0040  46 64 c0 f0 01 b0 00 25 3a 5b 49 ee c2 5e f8 da  --F...X...I...
0050  ad b6 a6 49 cc b7 8e 4f ba a5 79 9e d1 89 49 7f  --I...O...Y...
0060  25 96 84 10 31 56 18                                --...IV..
```

Bottom Screenshot: ICMPv6 Capture

No.	Time	Source	Destination	Protocol	Length	Info
5	0.145530	fe80::401c:11ff:fe0...	2409:40c0:1050:b9b6::	ICMPv6	86	Neighbor Solicitation for 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 from 42:1c:11:0b:7b:8e
6	0.145577	2409:40c0:1050:b9b6::	fe80::401c:11ff:fe0...	ICMPv6	86	Neighbor Advertisement 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 (sol, ovr) is at a0:02:a5:6f:39:39
1213	19.344330	fe80::401c:11ff:fe0...	2409:40c0:1050:b9b6::	ICMPv6	86	Neighbor Solicitation for 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 from 42:1c:11:0b:7b:8e
1214	19.344520	2409:40c0:1050:b9b6::	fe80::401c:11ff:fe0...	ICMPv6	86	Neighbor Advertisement 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 (sol, ovr) is at a0:02:a5:6f:39:39
2129	33.939341	fe80::401c:11ff:fe0...	2409:40c0:1050:b9b6::	ICMPv6	86	Neighbor Solicitation for 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 from 42:1c:11:0b:7b:8e
2130	33.939415	2409:40c0:1050:b9b6::	fe80::401c:11ff:fe0...	ICMPv6	86	Neighbor Advertisement 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 (sol, ovr) is at a0:02:a5:6f:39:39
2614	52.529119	fe80::401c:11ff:fe0...	2409:40c0:1050:b9b6::	ICMPv6	86	Neighbor Solicitation for 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 from 42:1c:11:0b:7b:8e
2615	52.529193	2409:40c0:1050:b9b6::	fe80::401c:11ff:fe0...	ICMPv6	86	Neighbor Advertisement 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 (sol, ovr) is at a0:02:a5:6f:39:39
2940	68.911469	fe80::401c:11ff:fe0...	2409:40c0:1050:b9b6::	ICMPv6	86	Neighbor Solicitation for 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 from 42:1c:11:0b:7b:8e
2941	68.911533	2409:40c0:1050:b9b6::	fe80::401c:11ff:fe0...	ICMPv6	86	Neighbor Advertisement 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 (sol, ovr) is at a0:02:a5:6f:39:39
3320	84.527451	fe80::401c:11ff:fe0...	2409:40c0:1050:b9b6::	ICMPv6	86	Neighbor Solicitation for 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 from 42:1c:11:0b:7b:8e
3321	84.527503	2409:40c0:1050:b9b6::	fe80::401c:11ff:fe0...	ICMPv6	86	Neighbor Advertisement 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 (sol, ovr) is at a0:02:a5:6f:39:39
3636	109.767368	fe80::401c:11ff:fe0...	2409:40c0:1050:b9b6::	ICMPv6	86	Neighbor Solicitation for 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 from 42:1c:11:0b:7b:8e
3637	109.767440	2409:40c0:1050:b9b6::	fe80::401c:11ff:fe0...	ICMPv6	86	Neighbor Advertisement 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 (sol, ovr) is at a0:02:a5:6f:39:39
3642	114.177570	fe80::ca3d:59a3:c4a...	ff02::1:ff0b:7b8e	ICMPv6	86	Neighbor Solicitation for fe80::401c:11ff:fe0b:7b8e from a0:02:a5:6f:39:39
3643	114.181519	fe80::401c:11ff:fe0...	fe80::ca3d:59a3:c4a...	ICMPv6	86	Neighbor Advertisement fe80::401c:11ff:fe0b:7b8e (rtr, sol, ovr) is at 42:1c:11:0b:7b:8e
3659	119.189931	fe80::401c:11ff:fe0...	fe80::ca3d:59a3:c4a...	ICMPv6	86	Neighbor Solicitation for fe80::ca3d:59a3:c4a1:342a from 42:1c:11:0b:7b:8e
3660	119.190597	fe80::ca3d:59a3:c4a...	fe80::401c:11ff:fe0...	ICMPv6	86	Neighbor Advertisement fe80::ca3d:59a3:c4a1:342a (sol, ovr) is at a0:02:a5:6f:39:39
3704	159.316960	fe80::401c:11ff:fe0...	2409:40c0:1050:b9b6::	ICMPv6	86	Neighbor Solicitation for 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 from 42:1c:11:0b:7b:8e
3705	159.317038	2409:40c0:1050:b9b6::	fe80::401c:11ff:fe0...	ICMPv6	86	Neighbor Advertisement 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307 (sol, ovr) is at a0:02:a5:6f:39:39

Packet 3705 details:

- Frame 5: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{D66...}
- Ethernet II, Src: 42:1c:11:0b:7b:8e (42:1c:11:0b:7b:8e), Dst: Intel_6f:39:39 (a0:02:a5:6f:39:39)
- Internet Protocol Version 6, Src: fe80::401c:11ff:fe0b:7b8e, Dst: 2409:40c0:1050:b9b6:8ce2:46a4:60f6:1307
- Internet Control Message Protocol v6

Hex dump and ASCII view:

```
0000  a0 02 a5 6f 39 39 42 1c 11 0b 7b 8e 86 dd 60 00  --o99B...{....
0010  00 00 00 20 3a ff fe 80 00 00 00 00 00 40 1c  --...:..@...@
0020  11 ff fe 0b 7b 8e 24 09 40 c0 10 50 b9 b6 8c e2  --...$ @...P...
0030  46 a4 60 f6 13 07 01 01 f2 0a 00 00 00 24 00  --F...$...$...
0040  40 c0 10 50 b9 b6 8c e2 46 a4 60 f6 13 07 01 01  --@...P...F...
0050  42 1c 11 0b 7b 8e                                --B...{...
```